

ORIGINAL PILOT STUDY

Carbon Aware Machine Learning Model Selection Under Accuracy Constraints

A reproducible benchmark using three public classification datasets

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Abstract

This pilot study evaluates whether simple accuracy constraints can reduce estimated training energy without materially reducing classification performance. Logistic regression, random forest, histogram gradient boosting, and a multilayer perceptron were evaluated across the Breast Cancer Wisconsin, Wine, and Digits datasets using five stratified train-test splits. CPU process time was converted to an explicit 35 W allocated-power energy proxy and then to emissions under 100, 400, and 700 gCO₂e/kWh scenarios. Logistic regression achieved the highest mean accuracy on Breast Cancer, 0.976, and Wine, 0.987, while also using the least estimated energy. On Digits, histogram boosting achieved 0.975 accuracy; random forest remained within 0.5 percentage points at 0.972 and reduced the proxy energy estimate by 93.9 percent. The absolute emissions estimates are not hardware-metered measurements. The study's main result is therefore comparative: reporting compute alongside accuracy can change model selection.

Research Questions

- How large are the runtime and estimated-energy differences among common classifiers on small public datasets?
- Does an accuracy-within-0.5-percentage-points rule select a lower-energy model?
- How does grid carbon intensity affect absolute emissions while preserving model rankings?

Introduction

Machine-learning papers commonly optimize predictive quality while reporting little about computation. Training emissions depend on hardware, runtime, power draw, data-center efficiency, and electricity supply. Green AI research argues that efficiency should be reported as a first-class result [1-4]. This study tests a simple decision rule: among models within 0.5 percentage points of the best mean accuracy, choose the model with the lowest estimated training energy.

Related Work

Strubell and colleagues quantified the energy and policy implications of large neural-network training [1]. Green AI proposed efficiency as an evaluation criterion alongside accuracy [2]. Later work introduced systematic emissions reporting and tools for machine-learning experiments [3, 4]. Patterson and colleagues showed that model, processor, data center, and energy supply jointly determine emissions [5]. The present experiment applies these reporting ideas to smaller student-scale workloads.

Data and Methods

Public Benchmark Data

The experiment used three datasets distributed with scikit-learn: Breast Cancer Wisconsin Diagnostic with 569 samples and 30 features, Wine with 178 samples and 13 features, and handwritten Digits with 1,797 samples and 64 features. Each dataset was split into 75 percent training and 25 percent testing with stratification under five fixed random seeds.

Models and Measurement

The benchmark compared standardized logistic regression, a 400-tree single-process random forest, histogram gradient boosting, and a standardized multilayer perceptron with two hidden layers and early stopping. CPU process time and wall time were recorded for fitting only. Estimated training

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energy equalled CPU seconds multiplied by an assumed 35 W allocated power and divided by 3,600. Emissions equalled energy in kWh multiplied by grid intensity. This transparent proxy supports within-run comparisons but is not a substitute for hardware power telemetry.

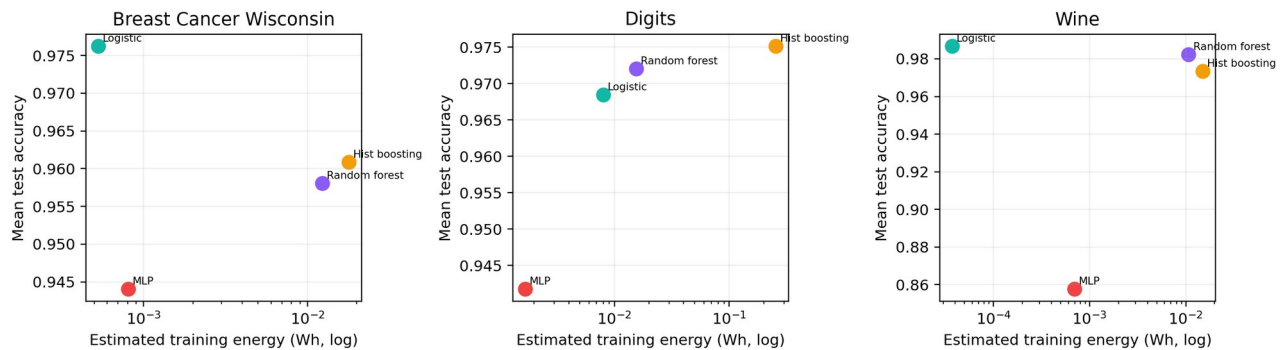
Decision Rule

Mean test accuracy and macro F1 were calculated across five splits. The carbon-aware candidate set contained every model within 0.005 absolute accuracy of the highest-scoring model for that dataset. The model with the lowest mean estimated energy in that set was selected. The rule was specified before interpreting the aggregated results.

Results

Dataset	Model	Accuracy	SD	Macro F1	CPU s	Est Wh
Breast Cancer Wisconsin	Histogram boosting	0.9608	0.0195	0.9573	1.8395	0.0179
Breast Cancer Wisconsin	Logistic regression	0.9762	0.0182	0.9742	0.0545	0.0005
Breast Cancer Wisconsin	MLP	0.9441	0.0171	0.9386	0.0830	0.0008
Breast Cancer Wisconsin	Random forest	0.9580	0.0192	0.9547	1.2692	0.0123
Digits	Histogram boosting	0.9751	0.0073	0.9749	26.1326	0.2541
Digits	Logistic regression	0.9684	0.0106	0.9684	0.8233	0.0080
Digits	MLP	0.9418	0.0206	0.9414	0.1720	0.0017
Digits	Random forest	0.9720	0.0040	0.9718	1.5967	0.0155
Wine	Histogram boosting	0.9733	0.0243	0.9745	1.5439	0.0150
Wine	Logistic regression	0.9867	0.0122	0.9860	0.0038	0.0000
Wine	MLP	0.8578	0.0433	0.8565	0.0713	0.0007
Wine	Random forest	0.9822	0.0186	0.9832	1.1011	0.0107

Accuracy-energy trade-offs across public datasets



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Figure 1 Mean test accuracy against estimated training energy. The horizontal position is an energy proxy based on CPU process time and a 35 W allocation, shown on a logarithmic scale.

Dataset	Best accuracy	Score	Carbon-aware choice	Score	Energy reduction %
Breast Cancer Wisconsin	Logistic regression	0.976	Logistic regression	0.976	0.000
Digits	Histogram boosting	0.975	Random forest	0.972	93.890
Wine	Logistic regression	0.987	Logistic regression	0.987	0.000

The decision rule did not trade away accuracy on Breast Cancer or Wine because logistic regression was both the most accurate and the lowest-energy eligible model. On Digits, random forest was 0.31 percentage points below histogram boosting and reduced the proxy energy estimate by 93.9 percent. Logistic regression used still less energy on Digits but fell 0.67 percentage points below the top model, outside the prespecified tolerance.

Carbon Intensity Scenarios

Grid intensity changes the absolute emissions estimate linearly. For example, the mean Digits histogram-boosting run used an estimated 0.2541 Wh, corresponding to approximately 0.0254, 0.1016, or 0.1778 gCO₂e at 100, 400, or 700 gCO₂e/kWh. These small values reflect the scale of the datasets and the fit-only measurement boundary. Larger models, repeated searches, and full pipelines would multiply the footprint.

Discussion

The experiment shows that the highest-accuracy model is not always the only defensible choice. On Digits, a small accuracy tolerance admitted a much lower-energy alternative. On two smaller datasets, the simplest linear model dominated both objectives. Carbon-aware selection should therefore begin with efficient baselines, report the measurement boundary, and avoid extensive hyperparameter search unless the expected accuracy gain justifies the additional compute.

Limitations and Research Integrity

- Energy values are estimates from CPU process time and an assumed 35 W allocation; they are not readings from a power meter.
- The virtualized CPU environment, library versions, thread scheduling, and background load may affect timing.
- The benchmark uses small tabular and image datasets and does not represent large-model training or accelerator workloads.
- The 0.5-percentage-point tolerance is a declared decision rule, not a universal standard.
- Medical benchmark performance does not establish clinical usefulness or safety.

Conclusion

Across three public datasets, simple models often matched or exceeded more computationally expensive alternatives. A prespecified tolerance reduced the Digits energy proxy by 93.9 percent with a 0.31-percentage-point accuracy reduction. The finding supports routine reporting of runtime, estimated energy, carbon-intensity assumptions, and accuracy-energy trade-offs.

Data and Code Availability

The computational package accompanying this manuscript contains the exact Python script, cached source data, derived features, result tables, and figures used in the analysis. The raw climate observations were requested from the NASA POWER Daily API. The machine-learning benchmark datasets were loaded through scikit-learn without manual modification.

Component	Included files
Code	run_experiments.py
Data	scikit-learn Breast Cancer Wisconsin, Wine, and Digits datasets
Results	paper3_raw_runs.csv, paper3_summary.csv, paper3_decisions.csv, and Figure 1

References

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